

ECEN 454 - Lab Report

Lab Number: 3

Lab Title: Cell Characterization using Spectre

Section Number: 341

Student's Name: Abhishek Bhattacharyya

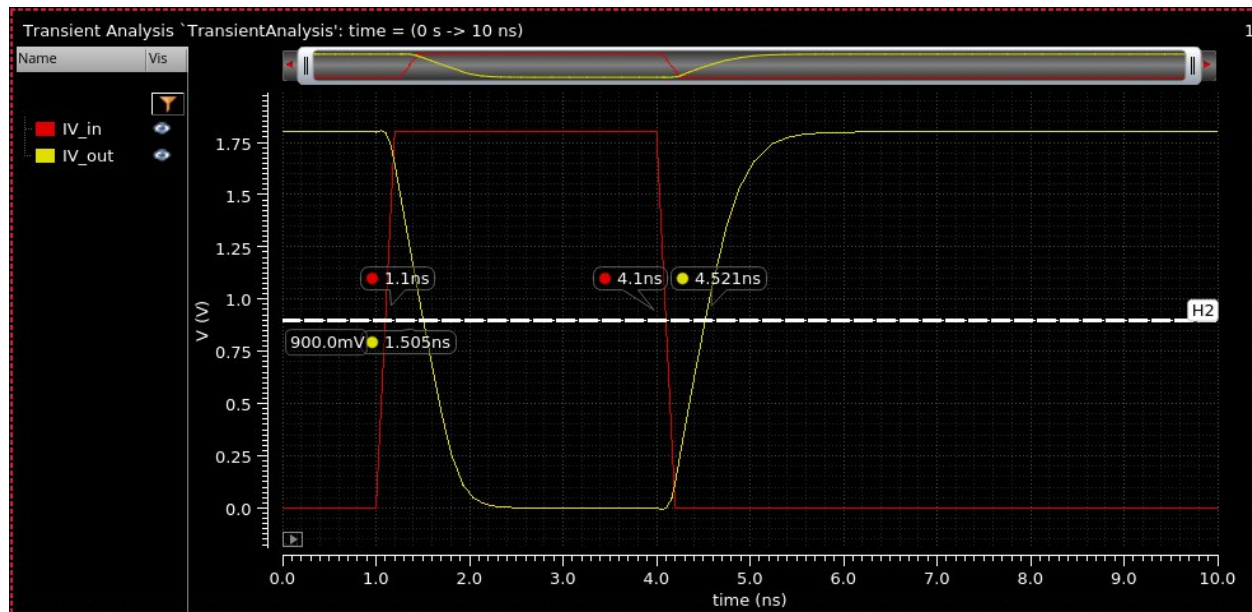
Student's UIN: 731002289

Date: 9/25/2024

TA: Nishank Bansal

Inverter

Transient Analysis Waveform



Parameters Used: $w_p = 1.1u$, $w_n = 0.4u$

Falling Delay: $1.505 - 1.1 = 0.405$ ns

Rising Delay: $4.521 - 4.1 = 0.421$ ns

Error Percentage: $(\text{Abs}(\text{Rising} - \text{Falling}) / \text{Min}) * 100 = 3.95\% < 10\%$ ✓

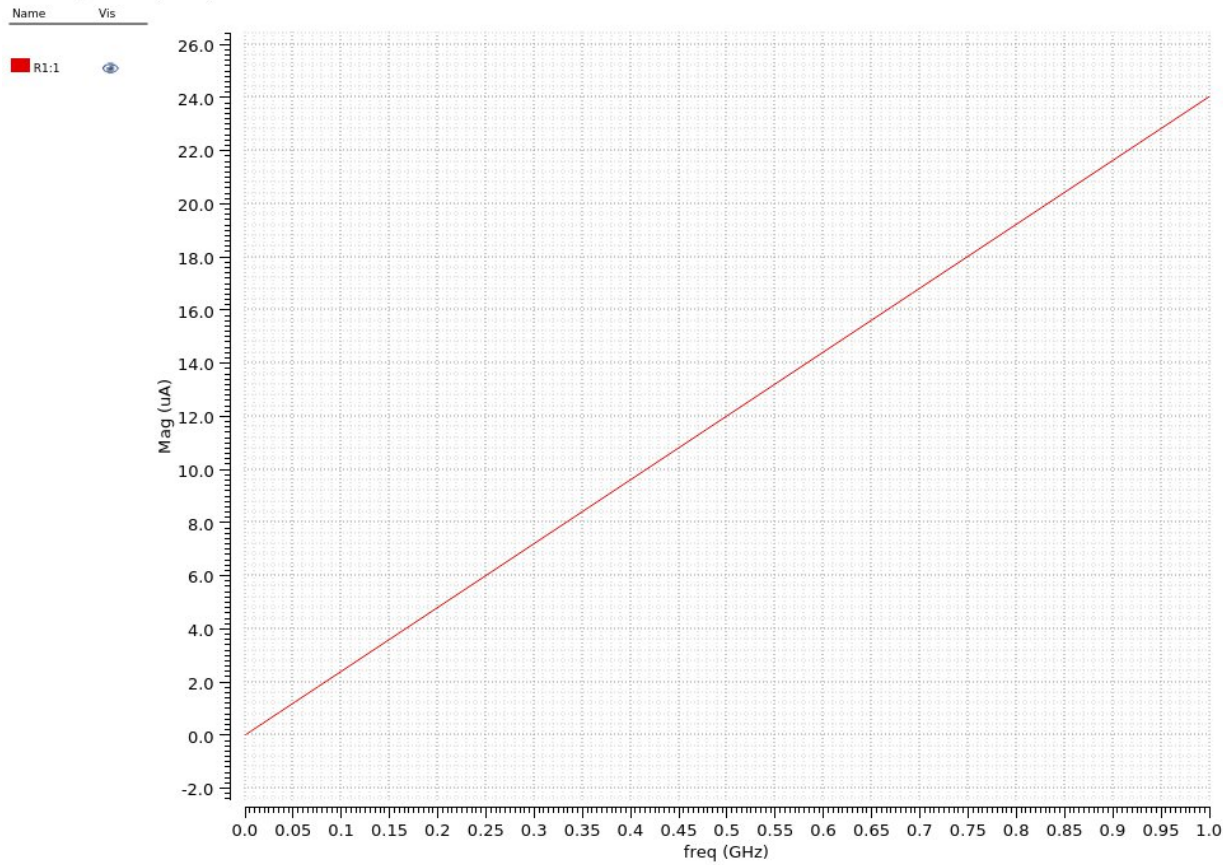
Delay Table

Capacitance (fF)	Falling Delay (ns)	Rising Delay (ns)	Error Percentage (green for <10%)
1	0.02664	0.04348	63.21%
8	0.06883	0.08367	21.56%
15	0.09868	0.11105	12.54%
22	0.12401	0.13635	9.95%
29	0.14936	0.16137	8.04%
36	0.17476	0.18683	6.91%
43	0.20001	0.21269	6.34%
50	0.22496	0.23734	5.50%
57	0.22496	0.23734	5.50%
64	0.24993	0.26438	5.78%
71	0.27545	0.29086	5.59%
78	0.32581	0.34258	5.15%
85	0.35251	0.36	2.12%
92	0.37763	0.39347	4.19%
100	0.4014	0.42049	4.76%
wn = 0.4		wp = 1.1	

AC Sim Waveform

AC Analysis `Freq`: freq = (10 Hz -> 1 GHz)

1



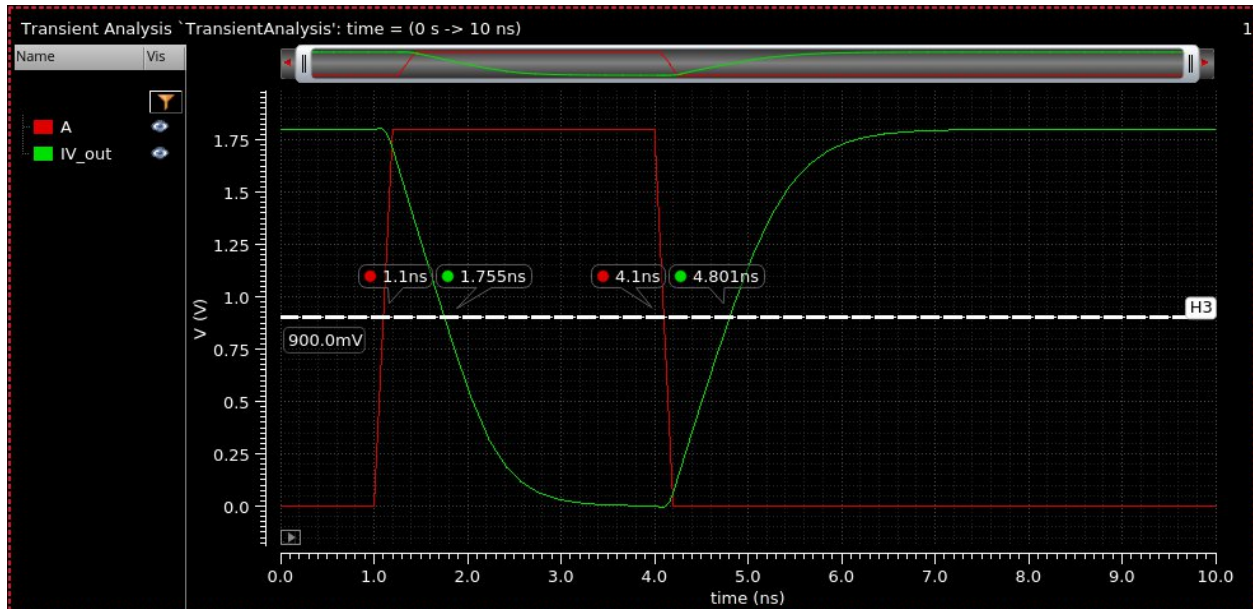
Sink Capacitance

Frequency (f)	Current (I)	Sink Capacitance
150	3.70E-12	3.92E-15
1.80E+08	4.32E-06	3.83E-15
3.04E+08	7.31E-06	3.83E-15
3.92E+08	9.42E-06	3.83E-15
4.81E+08	1.16E-05	3.84E-15
5.75E+08	1.38E-05	3.82E-15
6.79E+08	1.63E-05	3.82E-15
7.94E+08	1.91E-05	3.83E-15
9.17E+08	2.21E-05	3.84E-15
1.00E+09	2.40E-05	3.82E-15
Avg		3.84E-15

Therefore, mean sink capacitance is 3.84 femtofarad.

NAND

Transient Analysis Waveform



Parameters Used: $w_p = 0.6\mu$, $w_n = 0.3\mu$

Rising Delay: $1.755 - 1.1 = 0.655 \text{ ns}$

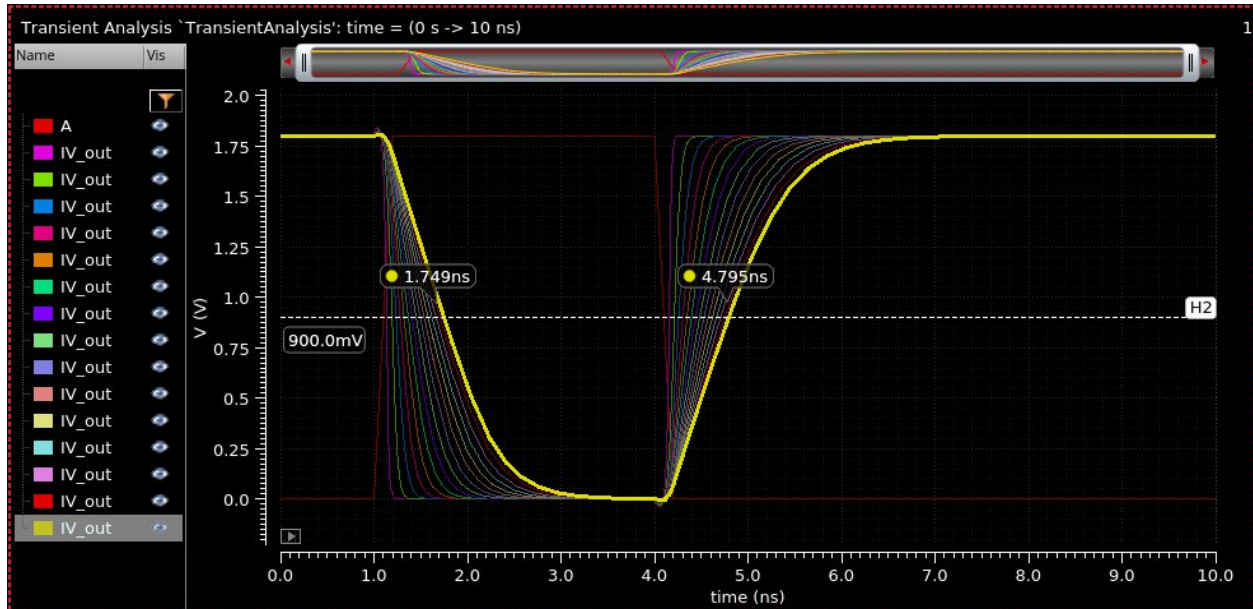
Falling Delay: $4.801 - 4.1 = 0.701 \text{ ns}$

Error Percentage: $(\text{Abs}(\text{Rising} - \text{Falling}) / \text{Min}) * 100 = 7.02\% < 10\% \checkmark$

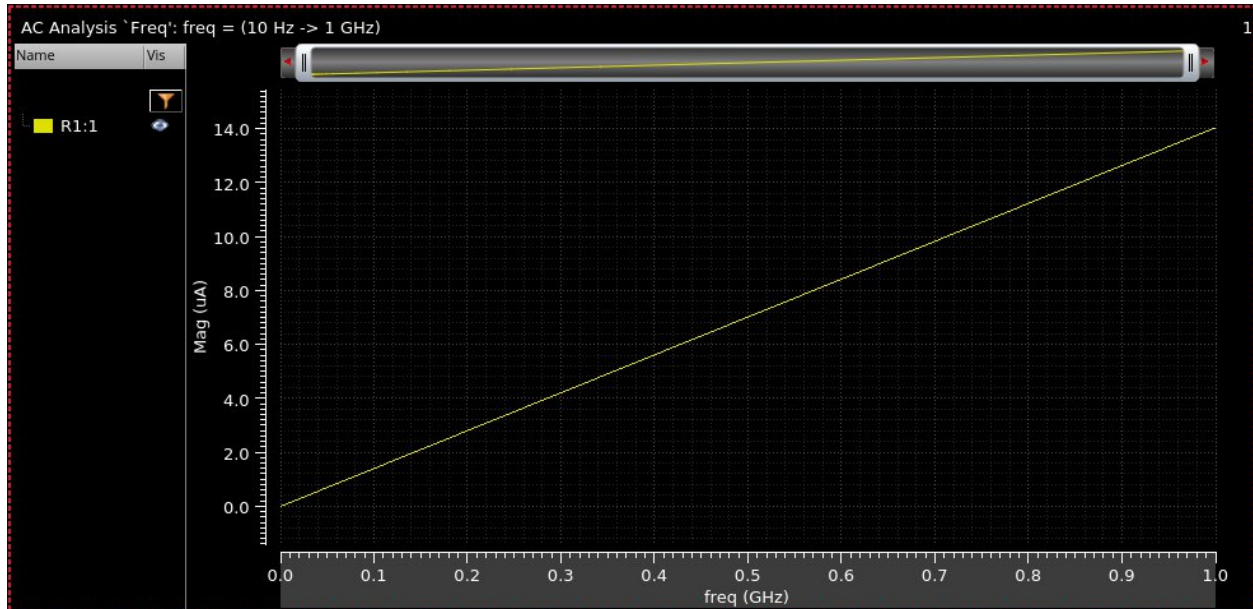
Delay Table

Capacitance (fF)	out1	out2	Falling Delay (ns)	Rising Delay (ns)	Error Percentage (green for <10%)
1	1.136	4.158	0.036	0.058	61.11%
8	1.194	4.213	0.094	0.113	20.21%
15	1.237	4.257	0.137	0.157	14.60%
22	1.28	4.303	0.18	0.203	12.78%
29	1.322	4.347	0.222	0.247	11.26%
36	1.366	4.394	0.266	0.294	10.53%
43	1.408	4.439	0.308	0.339	10.06%
50	1.451	4.482	0.351	0.382	8.83%
57	1.496	4.527	0.396	0.427	7.83%
64	1.537	4.573	0.437	0.473	8.24%
71	1.579	4.619	0.479	0.519	8.35%
78	1.621	4.661	0.521	0.561	7.68%
85	1.664	4.706	0.564	0.606	7.45%
92	1.707	4.749	0.607	0.649	6.92%
99	1.749	4.795	0.649	0.695	7.09%
w_n = 0.3				w_p = 0.6	

Waveform



AC Sim Waveform

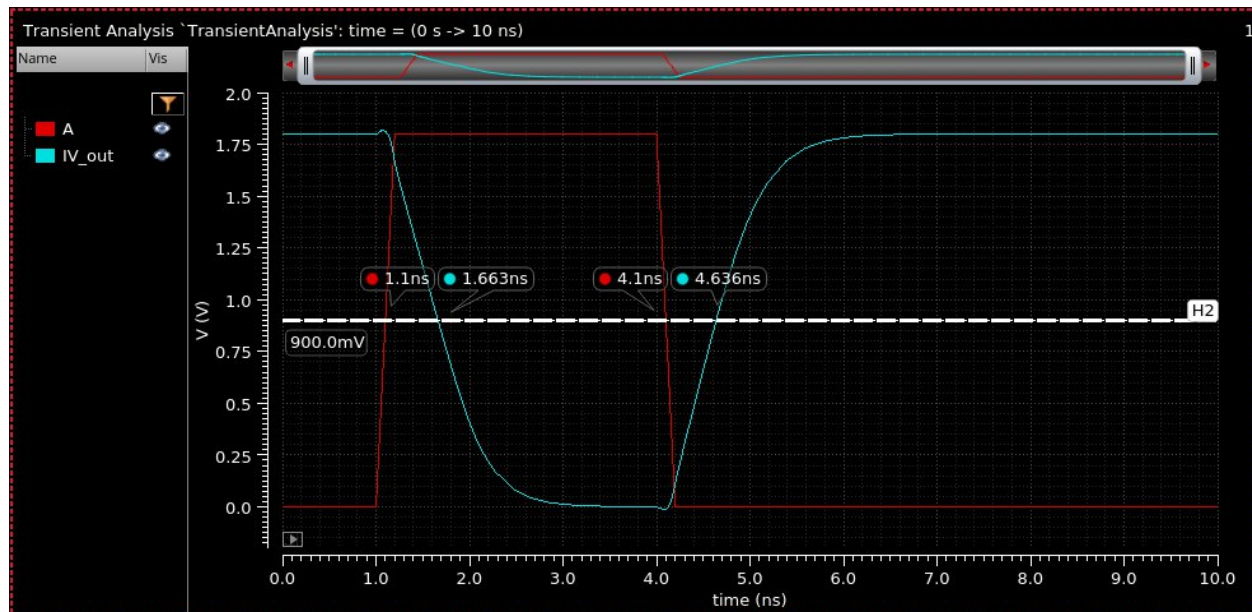


Sink Capacitance

Frequency (f)	Current (I)	Sink Capacitance
1.00E+08	1.41E-06	2.24E-15
2.00E+08	2.81E-06	2.24E-15
3.00E+08	4.21E-06	2.23E-15
4.00E+08	5.62E-06	2.24E-15
5.00E+08	7.02E-06	2.23E-15
6.00E+08	8.43E-06	2.24E-15
7.00E+08	9.83E-06	2.24E-15
8.00E+08	1.12E-05	2.24E-15
9.00E+08	1.26E-05	2.23E-15
1.00E+09	1.40E-05	2.24E-15
	Avg	2.23E-15

XOR

Transient Analysis Waveform



Parameters Used: $w_p = 1.6u$, $w_n = 0.4u$

Rising Delay: $1.663 - 1.1 = 0.563$ ns

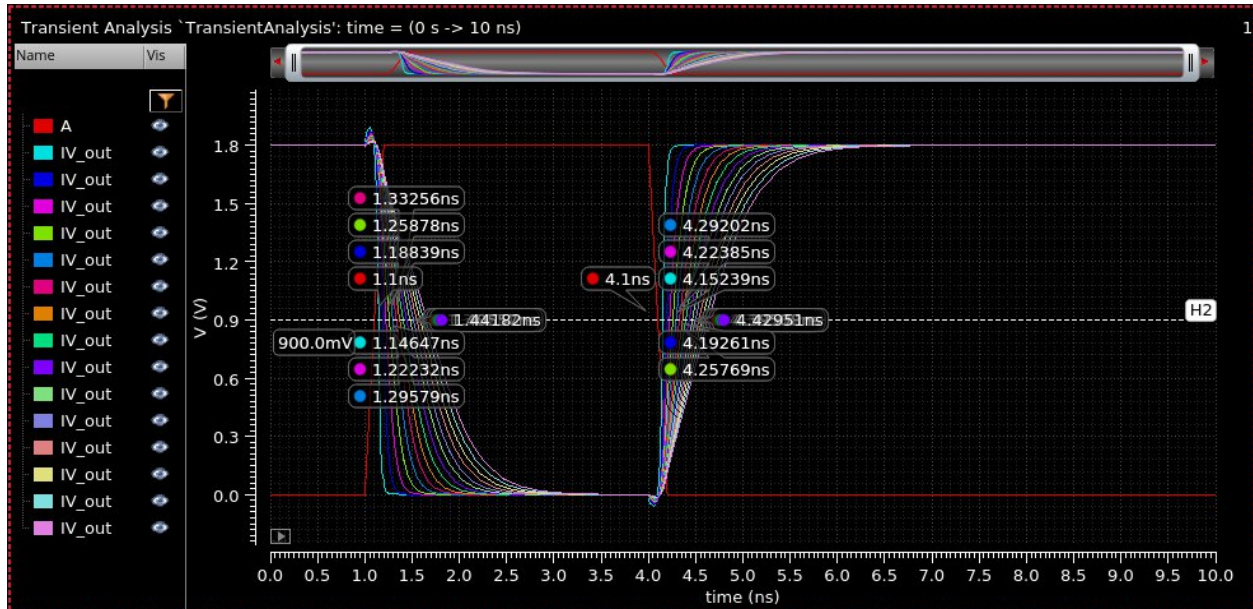
Falling Delay: $4.636 - 4.1 = 0.536$ ns

Error Percentage: $(\text{Abs}(\text{Rising} - \text{Falling}) / \text{Min}) * 100 = 5.04\% < 10\%$ ✓

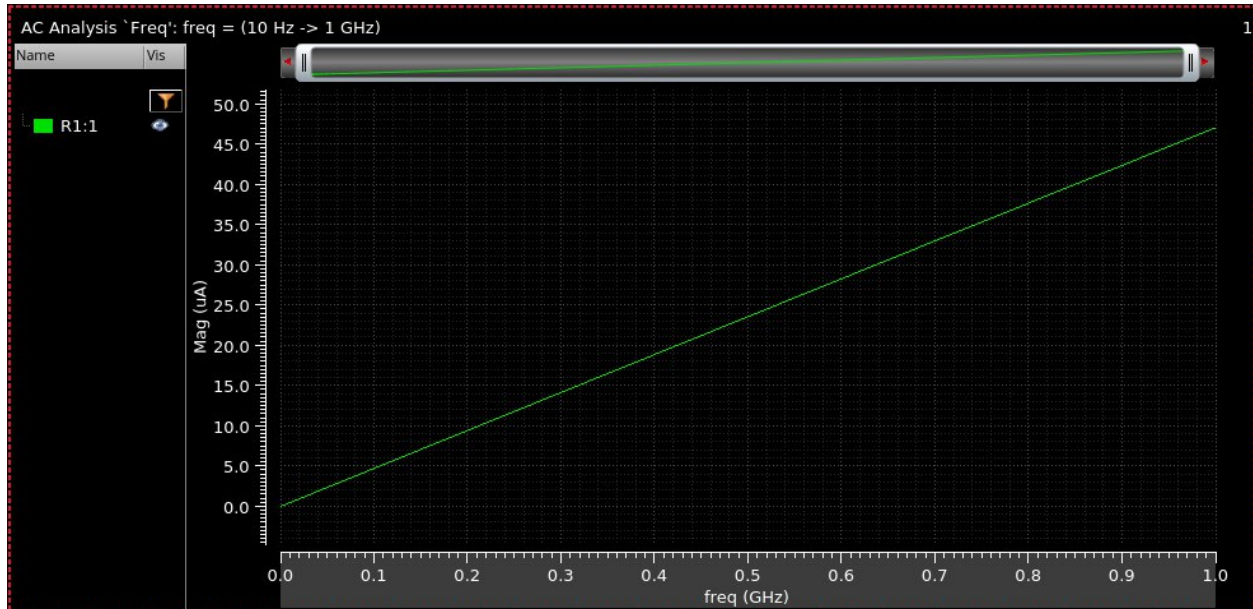
Delay Table

Capacitance (fF)	out1	out2	Falling Delay (ns)	Rising Delay (ns)	Error Percentage (green for <10%)
1	1.146	4.152	0.046	0.052	13.04%
8	1.188	4.193	0.088	0.093	5.68%
15	1.222	4.224	0.122	0.124	1.64%
22	1.259	4.258	0.159	0.158	0.63%
29	1.296	4.292	0.196	0.192	2.08%
36	1.333	4.326	0.233	0.226	3.10%
43	1.369	4.361	0.269	0.261	3.07%
50	1.406	4.395	0.306	0.295	3.73%
57	1.442	4.43	0.342	0.33	3.64%
64	1.478	4.462	0.378	0.362	4.42%
71	1.514	4.496	0.414	0.396	4.55%
78	1.551	4.532	0.451	0.432	4.40%
85	1.586	4.565	0.486	0.465	4.52%
92	1.622	4.598	0.522	0.498	4.82%
99	1.658	4.632	0.558	0.532	4.89%
w_n = 0.4				w_p = 1.6	

Waveform



AC Sim Waveform



Sink Capacitance

Frequency (f)	Current (I)	Sink Capacitance
1.00E+08	4.72E-06	7.51E-15
2.00E+08	9.44E-06	7.51E-15
3.00E+08	1.42E-05	7.51E-15
4.00E+08	1.89E-05	7.51E-15
5.00E+08	2.36E-05	7.51E-15
6.00E+08	2.83E-05	7.50E-15
7.00E+08	3.30E-05	7.50E-15
8.00E+08	3.77E-05	7.50E-15
9.00E+08	4.24E-05	7.49E-15
1.00E+09	4.71E-05	7.49E-15
	Avg	7.51E-15

Summary of Parameters

	W_P	W_N
Inverter	1.1u	0.4u
NAND	0.6u	0.3u
XOR	1.6u	0.4u

Files

Cell18.spi

```
//Spice netlist for an inverter
simulator lang=spectre
subckt IV (input output VDD VSS)
    parameters wp=0.6u lp=0.2u wn=0.3u ln=0.3u
    M1 output input VDD VDD tsmc18P w=wp l=lp
    M2 output input VSS VSS tsmc18N w=wn l=ln
ends IV
subckt NAND2 (A B out VDD GND)
    parameters wp=0.6u lp=0.2u wn=0.3u ln=0.3u
    M3 out B VDD VDD tsmc18P w=wp l=lp
    M2 out A VDD VDD tsmc18P w=wp l=lp
    M0 net2 B GND GND tsmc18N w=wn l=ln
    M1 out A net2 GND tsmc18N w=wn l=ln
ends NAND2
subckt XOR (A B out VDD GND)
    parameters wp=0.6u lp=0.2u wn=0.3u ln=0.3u
    M3 net3 net1 VDD VDD tsmc18P w=wp l=lp
    M2 out A net3 VDD tsmc18P w=wp l=lp
    M1 out net6 net2 VDD tsmc18P w=wp l=lp
    M0 net2 B VDD VDD tsmc18P w=wp l=lp
    M7 net5 B GND GND tsmc18N w=wn l=ln
    M6 net4 net1 GND GND tsmc18N w=wn l=ln
    M5 out A net5 GND tsmc18N w=wn l=ln
    M4 out net6 net4 GND tsmc18N w=wn l=ln
    I1 B net1 VDD GND IV
    I0 A net6 VDD GND IV
ends XOR
```

inverter.spi

```
//Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ecen454/cellcharacs/model18.spi"
include "~/ecen454/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vdd (vdd 0) vsource dc=1.8

vpwl (IV_in 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]

X1 (IV_in IV_out vdd gnd) IV wp=1.1u lp=0.2u wn=0.4u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save IV_in IV_out
```


Nand2.spi

```
Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ecen454/cellcharacs/model18.spi"
include "~/ecen454/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdd (vdd 0) vsource dc=1.8

vpwl (A 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]

X1 (A vdd IV_out vdd gnd) NAND2 wp=0.6u lp=0.2u wn=0.3u ln=0.2u

R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=100f

TransientAnalysis tran start=0 stop=10ns step=1ps
save A IV_out
```

Xor.spi

```
Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ecen454/cellcharacs/model18.spi"
include "~/ecen454/cellcharacs/cell18.spi"

vgnd (gnd 0) vsource dc=0
vvdd (vdd 0) vsource dc=1.8

vpwl (A 0) vsource type=pwl wave=[0n 0 1n 0 1.2n 1.8 4n 1.8 4.2n 0]

X1 (A vdd IV_out vdd gnd) XOR wp=1.6u lp=0.2u wn=0.4u ln=0.2u

#R1 (IV_out 1) resistor r=1
C1 (1 0) capacitor c=${DELAY}f

TransientAnalysis tran start=0 stop=10ns step=1ps
save A B IV_out
```

simcap_inverter.spi

```
Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ecen454_714/cellcharacs/model18.spi"
include "~/ecen454_714/cellcharacs/cell18.spi"

vgnnd (gnd 0) vsource dc=0
vdd (vdd 0) vsource dc=1.8

acinput (IV_in 0) vsource dc=0 mag=1

R1 (IV_in IV_in1) resistor r=0

X1 (IV_in1 IV_out vdd gnd) IV wp=0.9u lp=0.2u wn=0.4u ln=0.2u

Freq ac start=1e+1 stop=1e+9
save R1:currents
```

simcap_nand2.spi

```
Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ecen454/cellcharacs/model18.spi"
include "~/ecen454/cellcharacs/cell18.spi"

vgnnd (gnd 0) vsource dc=0
vdd (vdd 0) vsource dc=1.8

acinput (IV_in 0) vsource dc=0 mag=1

R1 (IV_in IV_in1) resistor r=0

X1 (IV_in1 vdd IV_out vdd gnd) NAND2 wp=0.6u lp=0.2u wn=0.3u ln=0.2u

Freq ac start=1e+1 stop=1e+9
save R1:currents
```

simcap_xor.spi

```
Spice netlist for an inverter and a capacitor
simulator lang=spectre

include "~/ecen454/cellcharacs/model18.spi"
include "~/ecen454/cellcharacs/cell18.spi"

vgnnd (gnd 0) vsource dc=0
vdd (vdd 0) vsource dc=1.8

acinput (IV_in 0) vsource dc=0 mag=1

R1 (IV_in IV_in1) resistor r=0

X1 (IV_in1 vdd IV_out vdd gnd) XOR wp=1.6u lp=0.2u wn=0.4u ln=0.2u

Freq ac start=1e+1 stop=1e+9
save R1:currents
```